

Selection Sort

Definition:

Selection Sort is a simple comparison-based sorting algorithm. It repeatedly finds the **smallest element** from the unsorted portion of an array and places it at the beginning of that portion.

Algorithm

1. Start from the first element.
2. Assume it is the minimum element.
3. Compare it with all remaining unsorted elements.
4. Find the actual smallest element.
5. Swap the smallest element with the first element of the unsorted part.
6. Move to the next position.
7. Repeat until the entire array is sorted.

Example

Array: **[64, 25, 12, 22, 11]**

- Pass 1 → smallest = 11 → **[11, 25, 12, 22, 64]**
- Pass 2 → smallest = 12 → **[11, 12, 25, 22, 64]**
- Pass 3 → smallest = 22 → **[11, 12, 22, 25, 64]**
- Pass 4 → smallest = 25 → **[11, 12, 22, 25, 64]**

Output:

```
Enter number of elements: 5
Enter the elements:
4
5
8
3
7
Sorted Array:
3 4 5 7 8
Execution Time: 4.0531158447265625e-05 seconds

Time Complexity:
Best Case   :  $O(n^2)$ 
Average Case:  $O(n^2)$ 
Worst Case  :  $O(n^2)$ 
Space Complexity:  $O(1)$ 

...Program finished with exit code 0
Press ENTER to exit console.
```

Selection Sort — Time and Space Complexity

Time Complexity

- **Best Case:** $O(n^2)$
- **Average Case:** $O(n^2)$
- **Worst Case:** $O(n^2)$

Space Complexity

- **Auxiliary Space:** $O(1)$
- It is an **in-place sorting algorithm**, so it does not require an extra array.